MILITARY SPECIFICATION

METER, SALINITY INDICATING, ELECTRIC

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 This specification covers one type of electric, salinity indicating meter with screw-stud terminals, suitable for use in alternating current, 60-cycle circuits and for flush mounting on a panel.

2. APPLICABLE DOCUMENTS

2.1 Issues of documents. The following documents of the issue in effect on date of invitation for bids or request for proposals, form a part of this specification to the extent specified herein:

SPECIFICATIONS

FEDERAL

PPP-B-636 - Boxes, Shipping, Fiberboard.

MILITARY

MIL-P-116 - Preservation-packaging, Methods of.

STANDARDS

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129 - Marking for Shipment and Storage.

(Copies of specifications, standards, drawings and publications required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Hqs., Defense Personnel Support Center, ATTN: Directorate of Medical Materiel, DPSC-ATT, 2800 S. 20th St., Phila., PA 19101, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

FSC 6630

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2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the effect on date of invitation for bids or request for proposals shall apply.

AMERICAN NATIONAL STANDARDS INSTITUTE (ANSI)

ANSI C39.1 Requirements for Electrical Analog Indicating Instruments

(Application for copies should be addressed to the American National Standards Institute, 1430 Broadway, New York, NY 10018.)

3. REQUIREMENTS

3.1 Material. Material shall be of the highest quality throughout and suitable in all respects for the purpose intended. The body shall be enclosed in a magnetically shielded metal case.

3.2 Style and design. The meter shall be round and suitable for flush mounting on salinity indicating panels for use in alternating-current, 60-cycle circuits. The meter shall have a full scale, legible without the use of external accessories. The meter shall conform to Figure 1.

3.2.1 Scale. The scale shall be nonlinear, calibrated in grains of sea salt per gallon. The scale shall be clockwise, 0 to 10 range, as shown in Figure 1. The area in the scale from 0 to 0.5 shall be a white band, from 0.5 to 1 a green band, and from 1 to 10 a red band.

3.2.2 Terminals. The meter shall have five screw-stud terminals, 0.187-32 thread, located and marked as shown in Figure 1.

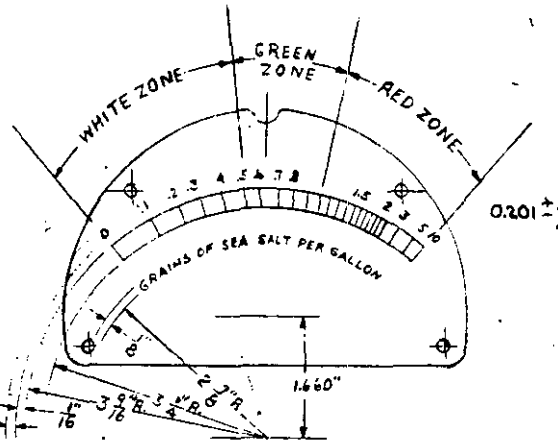
3.2.3 Dial markings. The dial shall have black scale markings (other than the specified color bands) on a white background.

3.2.4 Flange. Meter shall have a flange suitable for flush mounting of the meter as shown in Figure 1.

3.3 Dimensions. Dimensions of the meter shall be as shown in Figure 1 and as follows:

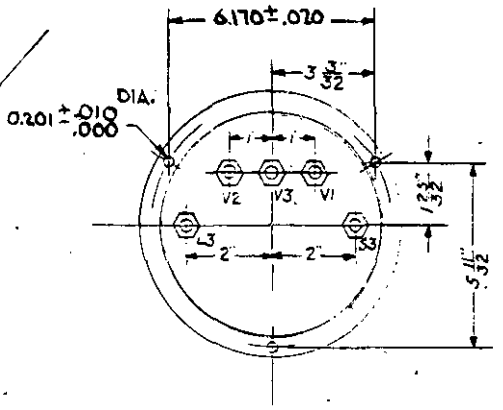
a. 3.575 ± 0.010 inches, depth of meter behind mounting flange, excluding terminals.
b. 7.65 ± .15 inches, diameter of mounting flange.
c. 6.25 ± .06, -.05 inches, diameter of meter body.
d. 4.575 ± .125 inches, overall depth of meter behind mounting flange, including terminals.

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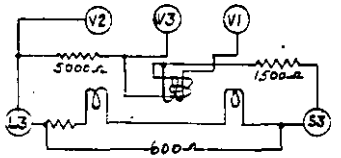


- NOTES -

1. DIAL TO BE CALIBRATED FROM TABLE OF RESISTANCE VALUES.
2. LENGTH OF SCALE TO BE NOT LESS THAN 4½ INCHES.
3. BLACK MARKINGS AND GRADUATIONS ON WHITE ENAMEL BACKGROUND WITH ZONES COLORED AS SHOWN.
4. WIDTH OF SCALE AND NUMERAL GRADUATIONS = .025 INCH. OTHER GRADUATIONS = .010 INCH.



TERMINAL LOCATIONS AND MOUNTING DIMENSIONS



WIRING DIAGRAM OF METER

(FOR STATOR COIL CIRCUIT ONLY)

(2) - ROTOR COILS - SAME AS POWER FACTOR METER - EACH COIL - 90 OHMS

(2) - STATOR COILS - WINDING, EACH COIL - 2200 TURNS OF #31 ENAMELED WIRE, (COPPER). RESISTANCE, EACH COIL - 135 - 185 OHMS, APPROX.

FIGURE 1. Meter, salinity, indicating, electric, flush mounting, round

METER CHECKING TABLE

THE FOLLOWING TABLE OF RESISTANCE VALUES IS USED FOR CALIBRATING THE DIAL AND CHECKING THE METER. WITH 60 VOLTS, 60 CYCLES CONNECTED ACROSS TERMINALS L3 AND S3, WITH A VARIABLE RESISTANCE "S" (SHOWN AT HEAD OF TABLE) ACROSS TERMINALS V2 AND V3 AND WITH ANOTHER VARIABLE RESISTANCE "C" (VALUES SHOWN IN TABLE OPPOSITE "SALINITY FIGURES") ACROSS TERMINALS V2 AND V1, THE POINTER INDICATION MUST BE WITHIN 0.078125 INCH OF THE SALINITY CORRESPONDING TO THE FOLLOWING TABLE.

TEST CONDUCTED AT ROOM TEMPERATURE OF 25° C.

SALINITY	50°F. TEST "S"=100,000 OHMS "C" AS SHOWN BELOW	110°F. TEST "S" = 4625 OHMS "C" AS SHOWN BELOW	200° F. TEST "S" = 1694 OHMS "C" AS SHOWN BELOW
0	INFINITY	INFINITY	INFINITY
.1	24,200	12,200	6,320
.2	14,100	7,200	3,680
.3	10,300	5,260	2,650
.4	8,000	4,080	2,060
.5	6,720	3,400	1,730
.6	5,720	2,910	1,470
.7	5,000	2,540	1,280
.8	4,450	2,260	1,140
.9	4,000	2,030	1,020
1.0	3,600	1,810	927
1.1	3,320	1,690	850
1.2	3.020	1,540	780
1.3	2,800	1,430	725
1.4	2,680	1,320	670
1.5	2,430	1,240	635
1.6	2,270	1,160	590
1.7	2,140	1,100	555
1.8	2,000	1,020	525
1.9	1,890	970	495
2.0	1,800	918	488
3.0	1,180	600	320
4.0	880	460	245
5.0	751	380	202
10.0	388	197	107

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3.4 Accuracy. When tested as specified in 4.5.1, the meter and dial shall be so calibrated that the pointer will indicate within 5/64 inch of the corresponding salinity graduation.

3.5 Magnetic influence. The meter shall meet the requirements specified on plate 15 of ANSI C39.1 for external-field influence when tested as specified in 4.5.2.

3.6 Working voltage. The meter shall meet the requirements specified on plate 15 of ANSI C39.1 for safe working voltage to ground when tested as specified in 4.5.3.

3.7 Identification marking. Each meter shall be permanently and legibly marked with the name or trademark of the manufacturer.

3.8 Workmanship. Workmanship shall be first class throughout. The meter shall be free from defects which detract from its appearance or may impair its serviceability.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Inspection. Inspection, as used in this specification, is defined as both examination (such as visual or auditory investigation without the use of special laboratory appliances or procedures) and testing (determination by technical means of physical and chemical properties) of the item.

4.1.2 Records. Records of examinations and tests performed by or for the contractor shall be maintained by the contractor and made available to the Government, upon the Government's request, at any time, or from time to time, during the performance of the contract and for a period of 3 years after delivery of the supplies to which such records relate.

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4.2 Sampling.

4.2.1 For examination. Sampling for examination shall be conducted in accordance with MIL-STD-105 and table I.

TABLE I. For examination

	Inspection level	AQL (percent defective
For visual examination		
Major defects	II	1.0
Minor defects	II	2.5
For dimensional examination	S-2	2.5

4.2.2 For tests. Sampling for tests shall be conducted in accordance with MIL-STD-105 and table II.

TABLE II. For tests

Component (unit of product)	Characteristic	Requirement	Test procedure	Inspection level	AQL
Meter	Accuracy	3.4	4.5.1	S-2	*
Meter	Magnetic influence	3.5	4.5.2	S-2	*
Meter	Working voltage	3.6	4.5.3	S-3	*

* Acceptance number shall be zero.

4.3 Examination. The meter shall be examined to determine compliance with all requirements contained in this specification.

4.3.1 Classification of defects. Examination shall be conducted in accordance with, but shall not be limited to, the defects listed in table III.

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TABLE III. Classification of defects

Category	Defect
Major	
101	Meter not free of cracks, pits, burrs and sharp edges.
102	Dial markings not complete, correct, legible and permanent.
103	Glass face not clear, and free of chips, blisters, checks or protrusions.
Minor	
201	Manufacturer's identification marking not present, complete, correct, legible and permanent.
202	Meter not free of dirt, grease or foreign material.

4.3.2 Dimensional examination. The meter shall be examined for defects in dimensions. Any dimension not within tolerance specified shall be classified as a defect.

4.4 Packaging inspection. The inspection of the unit, intermediate package, packing and marking for shipment and storage shall be in accordance with quality assurance provisions of the applicable container specification and the marking requirements of MIL-STD-129.

4.5 Tests. Tests shall be conducted to determine compliance with specification requirements.

4.5.1 Accuracy. The tests shall be performed in accordance with the instructions under Meter Checking Table (110° F. Test only) in Figure 1. The meter shall be tested in five positions: vertical, tipping instrument 45 degrees forward and backward from vertical and 45 degrees to left and right of vertical.

4.5.2 Magnetic influence. Meter shall be tested for external-field influence, as specified on plate 15 of ANSI C39.1.

4.5.3 Working voltage. Meter shall be tested for safe working voltage to ground, as specified on plate 15 of ANSI C39.1.

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5. PACKAGING

5.1 Preservation-packaging. Preservation-packaging shall be level A or C as specified (see 6.1).

5.1.1 Unit of issue. Each unit, as specified, constitutes one unit of issue.

5.1.2 Level A.

5.1.2.1 Cleaning. Meters shall be cleaned in accordance with MIL-P-116, process C-1.

5.1.2.2 Drying. Meters shall be dried in accordance with MIL-P-116.

5.1.2.3 Preservative application. Preservatives shall not be used.

5.1.2.4 Unit packaging. Meters shall be individually packaged in accordance with MIL-P-116, method IC-2.

5.1.3 Level C. Units shall be packaged in standard commercial containers of the size and kind commonly used, which will afford the degree of protection required for shipment and use of the product for its intended purpose.

5.2 Packing. Packing shall be level A, B or C, as specified (see 6.1).

5.2.1 Level A. Meters shall be packed in an exterior container designed for a type 2 load and conforming to PPP-B-636, class weather-resistant. Closure, strapping and waterproofing shall be as specified in the appendix of the box specification.

5.2.2 Level B. Meters shall be packed in an exterior container designed for a type 2 load and conforming to PPP-B-636, class domestic. Closure shall be as specified in the appendix of the box specification.

5.2.3 Level C. The subject commodity shall be packed in substantial commercial containers of the type, size and kind commonly used for the purpose, so constructed as to insure acceptance and safe delivery by common or other carriers, at the lowest rate, to point of delivery called for in the contract or purchase order.

5.3 Marking. Each unit package and exterior container shall be marked in accordance with MIL-STD-129.

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6. NOTES

6.1 Ordering data. Procurement documents should specify the following:

a. Title, number and date of this specification.
b. National Stock Number (NSN).
c. Selection of applicable levels of packaging and packing (see 5.1 and 5.2).

6.2 This specification covers the following item listed in the Federal Supply Catalog:

National Stock No.	Item Identification
6630-00-314-4407	METER, SALINITY INDICATING, ELECTRIC, Flush Mounting, Round

6.3 This specification does not cover all types, classes, grades or sizes of the commodity indicated by the title of this specification, or those which are commercially available, but is intended to cover types which are normally procured to meet military requirements.

Custodians:
Army - EA
Navy - MS
Air Force - 99

Preparing activity:
DSA-DM

Project No. 6630-0272

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

OMB Approval
No. 22-R255

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DOCUMENT IDENTIFIER AND TITLE

MIL-M-37448 - METER, SALINITY INDICATING, ELECTRIC

NAME OF ORGANIZATION AND ADDRESS	CONTRACT NUMBER
	MATERIAL PROCURED UNDER A ☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE DOCUMENT CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY DOCUMENT REQUIREMENT CONSIDERED TOO RIGID

3. IS THE DOCUMENT RESTRICTIVE?

☐ YES ☐ NO *(If "Yes", in what way?)*

4. REMARKS

SUBMITTED BY *(Printed or typed name and address - Optional)*	TELEPHONE NO.
	DATE

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